

COORDINATE - ENERGY TRANSFORMATION AND ONE-POINT FUNCTION FOR THE XXZ SPIN- $\frac{1}{2}$ CHAIN

NIKOLAUS ELSAESSER

JOINT WORK WITH

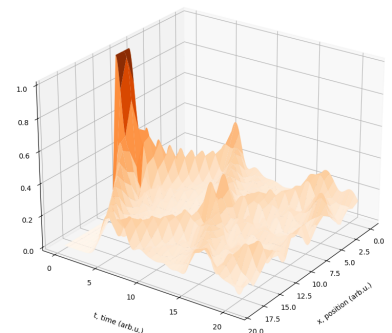
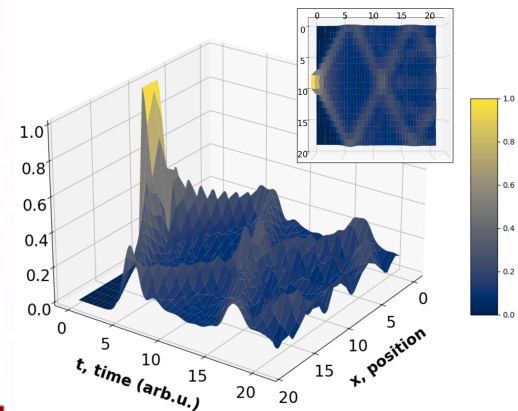
AXEL SAENZ

OREGON STATE UNIVERSITY

ERIC CORWIN

UNIVERSITY OF OREGON

One-point Probability Function ($\Delta = 0.04$)



CONTEXT & HISTORY (non-exhaustive)

- 1931 + Hans Bethe has ansatz for Heisenberg-Ising spin chain
- 1966 + Yang & Yang prove ground state w/ Bethe's ansatz
- ⋮
- Intense study of Bethe Ansatz methods & integrable systems from Leningrad, Kyoto, Canberra/Melbourne, Saclay schools (many more)

Cont'd, Interacting Particle Systems (IPS) § Integrable probability (IP)

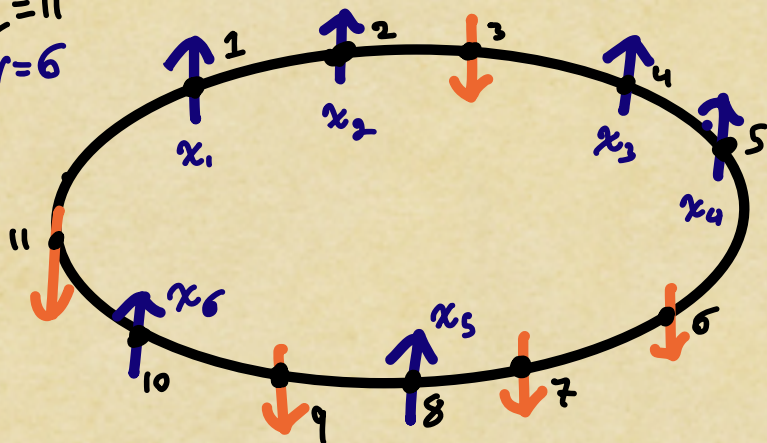
- 1992 + Gwa & Spohn - BA for ASEP on ring
 - 1997 + Schütz - BA for (T)ASEP on $\mathbb{Z}$
 - 2008 + Tracy & Widom - BA for ASEP on $\mathbb{Z}$,
exact results on one-pt function
 - 2015 + Borodin, I. Corwin, Petrov, Sasamoto
- Spectral theory for IPS solvable
by coord. BA
 - '16, '18, '19 + Baik & Liu
 - 2020 + Liu, Seitz, Wang
 - 2022 + Seitz, Tracy, Widom - BA for XXZ on $\mathbb{Z}$
- } BA for (T)ASEP,
q-TAZRP

* One-pt for XXZ $\rightarrow$ Evidence for KPZ universality (?)

Coordinate Bethe Ansatz Approach

XXZ on ring ($\mathbb{Z}/L\mathbb{Z}$)

$L=11$
 $N=6$



Coordinate basis

$$x \in \{(x_1, x_2, \dots, x_N)\}$$

e.g. $|\uparrow\uparrow\downarrow\uparrow\uparrow\downarrow\downarrow\uparrow\downarrow\uparrow\downarrow\rangle$
 $\equiv |1, 2, 4, 5, 8, 10\rangle$

$$\begin{cases} i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle \\ |\psi(0)\rangle = |y\rangle \end{cases}$$

$$H: |\dots \uparrow \downarrow \dots\rangle \mapsto \Delta |\dots \uparrow \downarrow \dots\rangle - |\dots \downarrow \uparrow \dots\rangle$$

BA: $|x\rangle \mapsto |z\rangle, \quad z \in \mathbb{C}^N$

$$|z\rangle = \sum_{\substack{\text{configs} \\ x}} u(z, x) |x\rangle$$

so that

$$H |z\rangle = E |z\rangle$$

Call $z = (z_1, \dots, z_N)$ Bethe vectors or energy basis

Long-Standing Concerns

- Does the Bethe Ansatz produce all the eigenvectors of $\mathcal{H}$?
i.e. Completeness
- Periodicity of $\mathbb{Z}/L\mathbb{Z}$ creates a consistency condition on Bethe vectors
a.k.a. the 'Bethe Equations'
 - Enumerating & finding solutions is non-trivial task

$$\forall i=1, \dots, N$$
$$\lambda_i^L = (-1)^{N-1} \prod_{j=1}^N \frac{1 + \lambda_i \lambda_j - 2\delta_{ij}}{1 + \lambda_i \lambda_j - 2\delta_{ij}}$$

Our Contribution

Energy to Coordinate transformation $|\lambda\rangle \mapsto |x\rangle$

Conjecture: $|x\rangle = \sum_{\substack{\text{Bethe} \\ \text{vectors} \\ \lambda}} \mathcal{L}(x, \lambda) |\lambda\rangle$

Equivalently $\mathbb{1}(x=y) = \sum_{\substack{\text{Bethe} \\ \text{vectors} \\ \lambda}} \mathcal{L}(y, \lambda) \mathcal{U}(\lambda, x)$

- Proof for $\Delta=0$, $N=2$; generic Δ in progress (odd L only)
- Numerically verified ($L=81$; $N=2$, $L=17$; $N=8, \dots$) (small Δ)

Counting Solutions to Bethe Equations (Brattain, Do, Saenz '15)

+ Conjecture $\Rightarrow$ Completeness

$$\mathcal{U}(x, \lambda) = \sum_{\sigma \in S_N} A_{\sigma}(\lambda) \prod_{i=1}^N \lambda_{\sigma(i)}^{x_i}$$

$$\mathcal{L}(y, \lambda) = \frac{1}{\det(\Delta(\lambda))} \sum_{\sigma \in S_N} \left(A_{\sigma}(\lambda) \prod_{i=1}^N \lambda_{\sigma(i)}^{y_i} \right)^{-1}$$

One-Point Function

Assume conjecture is true, then

$$|\Psi(t)\rangle = \sum_x \left(\sum_z l(y, z) u(z, x) e^{-iEt} \right) |x\rangle$$

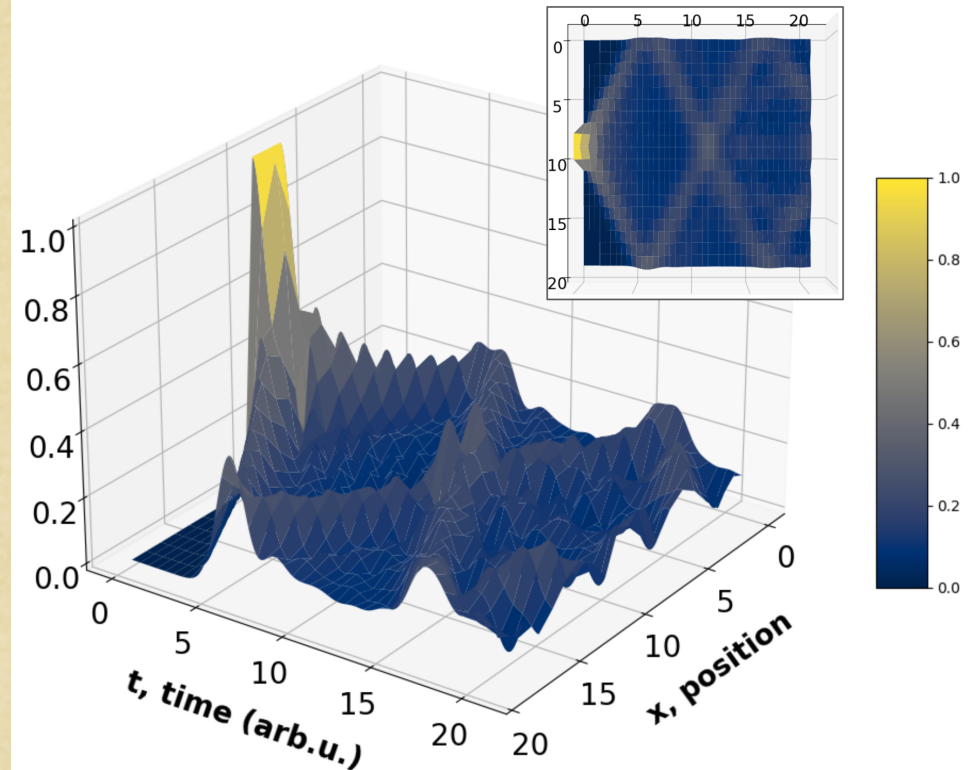
for $|\Psi(0)\rangle = |y\rangle$.

The probability of finding an up-spin particle at site $x_0 \in [L]$ then has the form

Theorem (E. Corwin - E. - Saeenz)

$$P(x_0, t) = \sum_{\substack{\text{Bethe vecs} \\ z, \bar{z}}} l(y, \bar{z}) F(x_0; \bar{z}, z) l(y, z)$$

One-point Probability Function ($\Delta = 0.04$)



$$N=3, L=21, |y\rangle = |8, 9, 10\rangle$$

THANK YOU!

Details can be found on ArXiv 

Coordinate-energy transformation and the
one-point function for the Heisenberg-Ising XXZ
spin- $\frac{1}{2}$ chain on the ring

Eric I. Corwin, Nikolaus Elsaesser, Axel Saenz

arXiv: 2506.14171

Revisions in progress: more literature review + $N=2$ proof